Integer-valued delivery promises: calibration, coverage and temporal shift

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1. Summary

A promised delivery date comes from three steps: a forecast, a calibration that sets its level, and a rule that turns it into whole days. In this study the last two steps moved the results as much as the choice of forecasting model did.

I started this project wanting to predict which Olist orders would arrive late. Along the way I realised that "late" compares two things: how long delivery takes, and how generous the promised date is. The question became: **when does a better forecast actually give a better promised date?** The data are public Olist e-commerce orders (Brazil, 2017–2018). The project's population keeps single-seller orders that were eventually delivered and whose Olist estimated delivery week falls on or before 23 July 2018; Section 3 explains why that last condition matters. A promise is a whole number of days; an order is covered if it arrives in time; the target coverage is 90%.

I compared a per-state quantile rule, a linear quantile regression and a shallow gradient-boosted model (HGB) under a protocol written down before any new model was fitted. Three findings:

1. **The comparison design changed the verdict.** In the project's earlier comparison, regression promised 2.57 more days than the state rule. With the same training data, the same calibration procedure and the same feature timing, it promised 1.54 fewer days at 0.31 points lower coverage.
2. **The frozen HGB and linear policies reached different coverage.** On the evaluation period the HGB policy covered 97.1% with 23.67 days and the linear policy 96.2% with 22.45 days. This compares two frozen policies at different realised coverage; it does not show which model is better at equal reliability. A post-hoc diagnostic that aligns coverage using the evaluation outcomes puts the HGB curve 0.6–0.9 days below the linear one, in this sample and for this family of shift policies.
3. **The registered policies did not meet the pre-set standard for replacing the existing rule.** That is a statement about these policies, not evidence that the models have no decision value: the standard gives no credit for extra coverage, and the outcome depends on the calibration level and on how the bound is rounded to whole days (Section 5, post hoc).

The evaluation orders had already been examined earlier in the project, so this is a pre-registered exploratory study, not a blind test. Every analysis added after the scoring run is labelled post hoc and changes no registered verdict.

2. Loss and decision: why a quantile, not the mean

If a late day and a spare day cost different amounts, the best promise is a quantile of the delivery time, not its mean. Let Y be the delivery time in whole days and d the promise, with the loss

$$L(d, Y) = c_{late} \cdot (Y - d)_+ + c_{slack} \cdot (d - Y)_+.$$

Let R(d) = E[L(d, Y)] and F the distribution function of Y. Moving the promise from d to d + 1 saves c_late on every order with Y > d and adds c_slack on every order with Y ⩽ d. Therefore

$$R(d + 1) - R(d) = (c_{late} + c_{slack}) \cdot F(d) - c_{late}.$$

This difference never decreases in d, so R is minimised at the smallest whole d with F(d) ⩾ q, where q = c_late / (c_late + c_slack). That is the empirical quantile rule the project uses (NumPy's inverted_cdf). The pinball loss at level q is this loss with c_late = q and c_slack = 1 − q (scikit-learn); a 90% target thus corresponds to a 9 : 1 penalty ratio. The mean minimises squared error and the median a 1 : 1 absolute loss; both treat a late day and a spare day alike.

A check on validation data. On the 12,045 validation orders I gave every order the same promise and computed the 9 : 1 loss for d = 1, …, 60. The minimum is at d = 20, the empirical 90% quantile: F(19) = 89.6% < 90% ⩽ F(20) = 91.4%. Promising the mean (11.43, rounded up to 12) would have covered only 65.5% of orders, at a loss of 23.76 per order against 15.61.

What 9 : 1 does not mean. It follows from choosing q = 0.90; I did not measure it, and it is not Olist's real cost ratio. Customer harm may not grow linearly with late days; if one late day already does most of the damage, a coverage constraint is closer to the truth than a linear cost. Therefore the study reports the trade-off between coverage and promised days. It does not claim a profit-optimal date.

3. Sample and comparison design

Sample. The population is defined on Olist's own estimated delivery week: an order enters only if that week lies between 27 February 2017 and 23 July 2018, its final outcome is delivered (on time or late), and it has a single seller. For the evaluation period E (purchases from 1 April 2018) this keeps 19,950 of 32,803 purchases. The week condition

removes 5,175 of the 6,292 July purchases and every later one, so a late-window order enters only if Olist promised it a short time. The evaluation sample is therefore filtered on one of the comparators (Supplement S3c).

Matched design. Each calibrated policy used the same training orders, the same label cut-off, the same rule for seller-history features and the same calibration procedure. A matched procedure is not automatically neutral: rounding the calibrated bound up treats a continuous forecast differently from a whole-day rule (Section 5). The uncalibrated controls (B1 and the retained rule B0) are kept apart. A label counts only if the order was delivered before the decision date.

Segment	Purchases	Orders	Late rate*	Role
F0 → V	Jan–Jun → Jul–Sep 2017	13,616 / 12,045	3.89% / 3.38%	fit and select settings
F	Jan–Sep 2017	25,666	3.67%	final fit
C	Oct–Dec 2017	17,056	8.74%	calibration shift
E	Apr–Jul 2018	19,950	4.26%	final scoring run

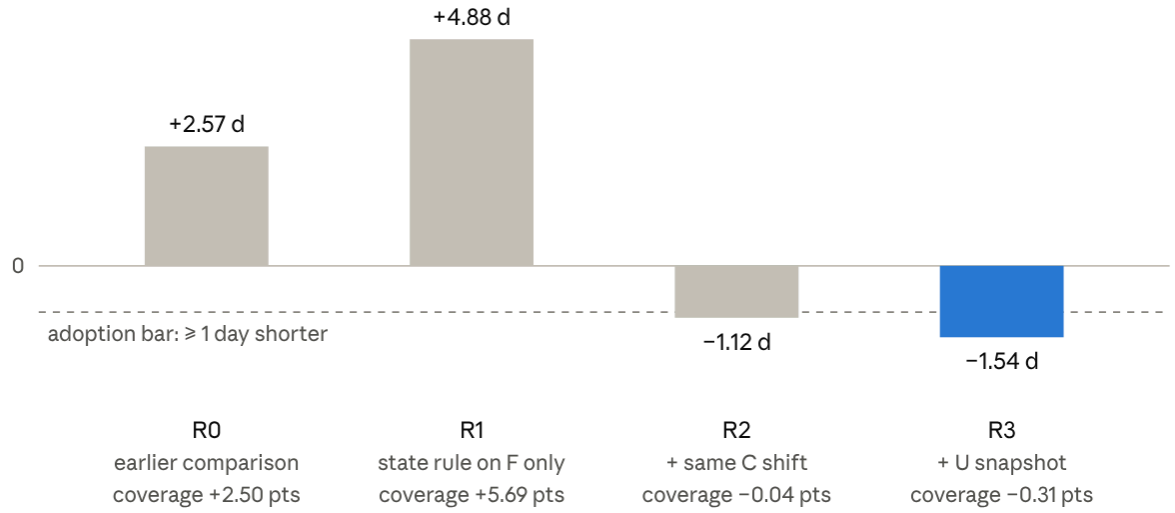
*Against Olist's own estimated date.

Calibration. Each calibrated policy is shifted by its own number computed on C: the k-th smallest of its residuals $y - \hat{y}$, with $k = \lceil (n + 1) \cdot 0.9 \rceil$ and $n = 17{,}056$. The promise is then rounded up: $\max\{1, \lceil \hat{y} + \text{shift} \rceil\}$. This has the form of split conformal, but its guarantee needs C and E to be exchangeable, and C had a much higher late rate than E (Barber et al.). Section 5 shows how this rounding treats continuous and whole-day predictors differently.

Stepwise comparison. The earlier comparison (R0) differed in three conditions at once, so I changed one condition per step. The pre-set primary comparison is R3 against R0; R1 and R2 are descriptive.

Matching the design flipped the gap from +2.57 to −1.54 days

Regression minus state rule, mean promised days on E (19,950 orders); one condition changed per step



outputs/e1/score/h1_steps.csv · E, April–July 2018, 19,950 orders

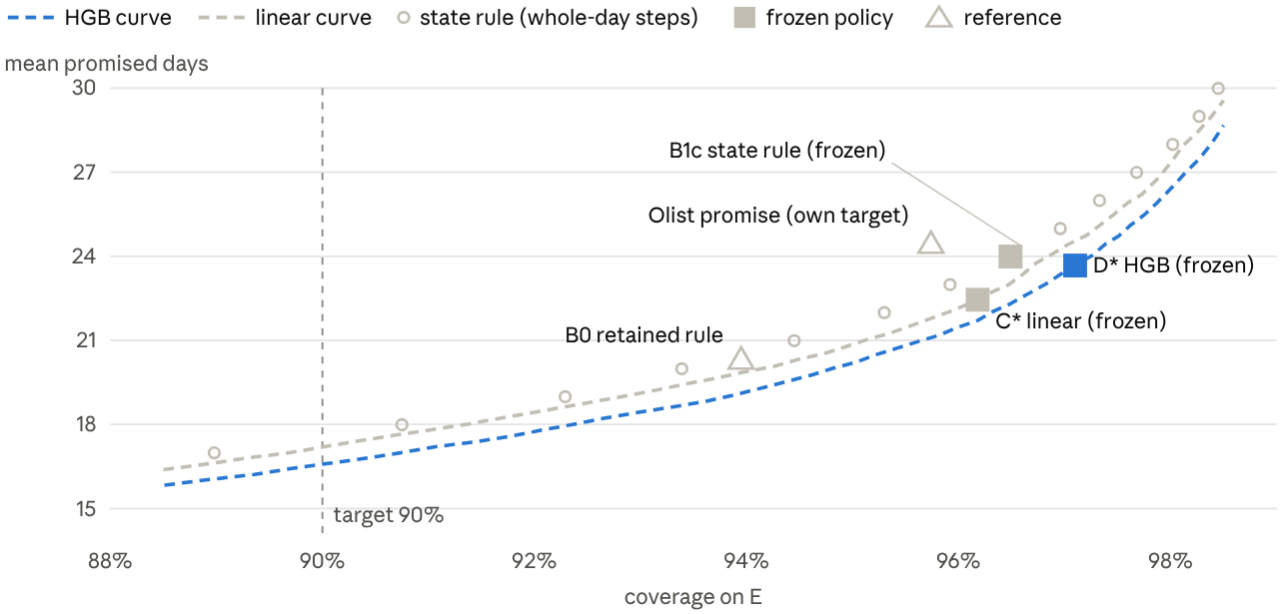
Under this fixed order, changing the design alone reversed the judgement; the largest step is giving the state rule the same calibration shift (+6 days). This is design sensitivity, not a causal statement about which method is better. Two references are kept apart: the matched reference B1c (registered under the name "fair reference": the state rule on F with the same calibration) and the replacement reference B0 (the project's retained rule, fitted on F + C without a shift).

4. Results

The E1 models and decision rules were frozen before the final scoring run on E, and each policy was scored once in that run. The figure keeps two kinds of evidence apart. Filled squares are those frozen policies, calibrated on C only. Dashed lines are diagnostic curves: each model's shift is moved using E's own outcomes, which aligns coverage after the fact. They describe the trade-off between days and coverage; they are not deployable policies and were not used to choose a model.

The frozen policies landed at different coverage on E

E, 19,950 orders. Filled squares: policies calibrated on C only, frozen, scored once on E.
Dashed lines: diagnostic curves whose shift uses E's own outcomes; descriptive only, not deployable, not used to choose a model.



outputs/e1/posthoc_m4/diagnostic_curves_E.csv and outputs/e1/score/policy_table.csv · E, 19,950 orders · dashed curves use E's outcomes, are post hoc and choose no policy

The frozen HGB policy covered 97.10% with 23.67 days, the frozen linear policy 96.18% with 22.45 days and B1c 96.49% with 23.99 days. Whole-day promises make coverage move in steps, so every diagnostic comparison reports the coverage actually reached rather than an interpolated match. At 90.02% HGB needs 16.60 days and the linear model 17.20; at 96.18% they need 21.71 and 22.44, while the state rule's nearest step is 96.49% at 23.99 days. In this sample and for this family of shift policies the HGB curve is the more favourable one; that is not a general ranking of the two models. On C at the registered shifts, HGB covered 91.14% with 24.54 days and the linear model 91.19% with 24.68.

The registered verdicts are unchanged. Intervals are week-block bootstrap intervals over 17 purchase weeks of unequal size with the models held fixed; they leave out refitting and selection uncertainty and are approximate. As a stability check (not a new interval), removing any one purchase week never changes the sign of a registered difference (Supplement S3e).

Question	Comparison	Registered result	Verdict
H1: does the matched design change the judgement?	R3 vs R0	gap moved by −4.11 days	met

Question	Comparison	Registered result	Verdict
H2: does a linear change help?	C* vs B3	V selected no new linear setting, so no final comparison between different linear models was formed	recorded as "not supported"
H3: does shallow HGB help?	D* vs B1c	−0.32 days (about −0.55 to −0.17), coverage +0.61 pts	below the 1-day bar
Replace the retained rule?	C*, D* vs B0	+2.15 and +3.37 days	standard not met

At q = 0.9126, a target matching Olist's coverage on C, the linear policy promised 23.42 days at 96.62% coverage against Olist's 24.44 days at 95.74%. It passes the pre-set rule by 0.02 days, and a post-hoc interval for the day difference (−3.25 to +0.69) includes zero. Section 5 adds a further reason for caution: the evaluation sample is filtered on Olist's own estimated date.

5. Rounding, sample window and calibration period

Four features of the design move the registered points. All numbers below are post hoc: they reuse the frozen predictions, choose no new policy and change no registered verdict.

Rounding to whole days. The registered promise rounds the calibrated bound $U = \hat{y}$ + shift up. Because Y is a whole number, $Y \leqslant U$ exactly when $Y \leqslant \lfloor U \rfloor$, so rounding down keeps the same coverage event; rounding up adds slack only when U is not a whole number, i.e. for continuous forecasts. Rounding down would shorten each continuous promise by exactly 1 day on E and leave B1c unchanged. Coverage on C would fall from 91.1–91.2% to 90.01%, which still meets the rank bound, and HGB would be 1.32 days shorter than B1c at +0.18 points. A plain "subtract one day" is not the same rule: it fails whenever U is a whole number (Supplement S3b). The registered rule is therefore conservative rather than wrong, and the registered H3 verdict stands.

Evaluation window. July contributes 1,085 orders, with Olist promises of at most 26 days (56–63 in April–June) and a mean actual time of 6.2 days. In April–May alone, coverage is 95.1% for the linear policy, 96.4% for HGB and 95.5% for B1c; in June–July it is 98.3–98.5%. The window explains part of the over-coverage, and the comparison with Olist is filtered on the comparator itself.

Calibration period. C runs from October to December 2017. Late rates were 4.2%, 12.5% and 7.4% by month, and November holds 42% of C, so a shift learned there may over-cover a calmer period. This is a candidate explanation; a high November late rate does not

by itself show that it caused the over-coverage.

Replacement standard. A policy may lose up to 1 point of coverage but gets no credit for extra coverage. At B0′s own coverage (93.95%) on E, the diagnostic curves need 19.12 days for HGB and 19.83 for the linear model, against 20.31 for B0. The verdict says that the registered policies missed the standard, not that the models lack decision value.

These four points do not overturn the registered results. They change how the results should be read: each registered difference mixes forecast quality with calibration level, rounding and sample window, and this study does not separate those parts.

6. Limits and conclusion

Limits. One evaluation period, already examined earlier in the project and filtered on Olist′s estimated week; a conditional population (single-seller, eventually delivered orders); a small stepwise model search; approximate intervals from 17 week clusters; no separate test of seasonality. Promised dates do not change logistics, so nothing here shows faster delivery or happier customers.

Conclusion. The pre-registered verdicts stand: the matched design reversed the regression-versus-rule judgement, V selected no new linear setting, HGB did not clear the 1-day bar, and the registered policies did not meet the replacement standard. The sensitivity checks narrow what these verdicts mean. The frozen policies reached different coverage, rounding up gave continuous forecasts extra slack, and the evaluation window was filtered on Olist′s own dates. The next step is a small, separately recorded methods study: how to turn a continuous calibrated bound into a whole-day promise (rounding down and its boundary cases), and a paired simulation that separates rounding, temporal shift and selective labels. Online recalibration is left for later, because Olist labels arrive late and the first orders to be delivered are not a random subset.

Supplementary material (model selection, the simulation, all post-hoc sensitivity tables), the integrity record and a contribution statement: Supplement (a separate PDF by email).